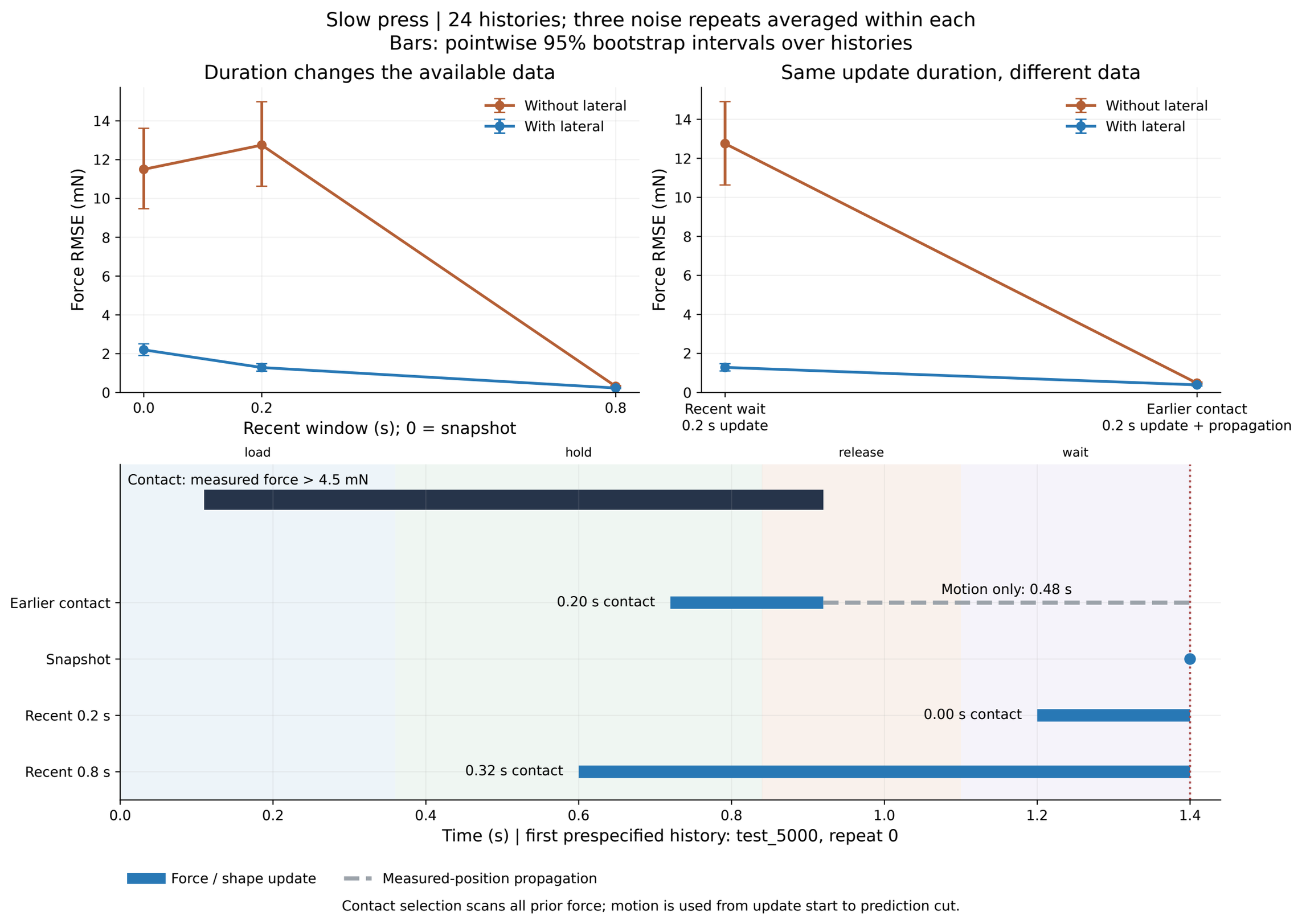

Slow press | 24 histories; three noise repeats averaged within each
Bars: pointwise 95% bootstrap intervals over histories
Duration changes the available data
Same update duration, different data
Without lateral
With lateral
Force RMSE (mN)
Recent window (s); 0 = snapshot
Recent wait
0.2 s update
Earlier contact
0.2 s update + propagation
load
hold
release
wait
Contact: measured force > 4.5 mN
Earlier contact
Snapshot
Recent 0.2 s
Recent 0.8 s
0.20 s contact
Motion only: 0.48 s
0.00 s contact
0.32 s contact
Time (s) | first prespecified history: test_5000, repeat 0
Force / shape update
Measured-position propagation
Contact selection scans all prior force; motion is used from update start to prediction cut.